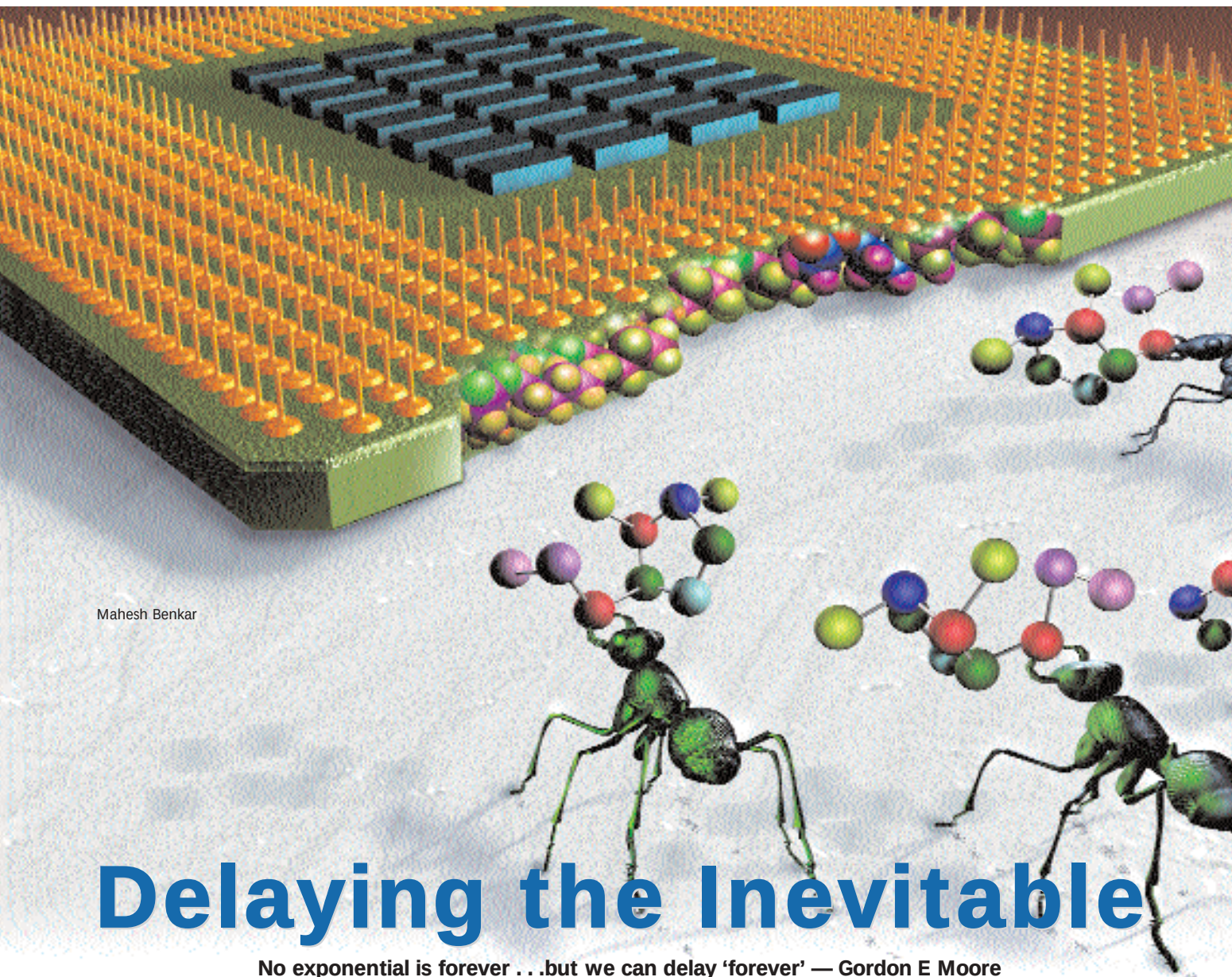




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Delaying the Inevitable

No exponential is forever . . .but we can delay 'forever' — Gordon E Moore

If necessity is the mother of invention, then the necessity for the semiconductor industry to keep inventing is Moore's Law. In an article published in *Electronics*, in 1965, Intel co-founder, Gordon Moore, predicted the rate at which the number of components on a single chip would double every two years. To understand the effect of Moore's law, consider this—Intel's first processor 4004, contained 2,300 transistors, while the current generation Itanium2 contains 410 million transistors.

So, what's the problem? This shrinking of chips cannot go on forever. Gordon Moore is optimistic that, at least for a decade, his law is safe. The general perception is that this law can't be extended after 2015 to 2020. The big question is, how long can we sustain it?

Chips are manufactured using a process known as lithography. The basic ingredient of integrated circuits is silicon in its purest form. This is obtained from its ore using a chemical process. It is

then shaped into circular discs called silicon wafers, the diameters of which vary depending on the manufacturing plant. The chips are then manufactured on these wafers using lithography. Layers built on top of the silicon wafer form the transistors, and connections between

them are etched using a powerful light beam. Multiple transistors when connected together form a chip. Many chips can be fabricated on a single silicon wafer. The percentage of chips produced without faults is indicated by the term 'yield'.

Adding transistors enables the chip to provide extra features such as a larger instruction set. As the transistors get smaller, the connections need to get thinner. Chips have already shrunk so much that some connections between the transistors are just a few atoms wide. Components in a chip are measured in nanometres—a billionth of a metre. Manufacturing them at such a scale is no easy task; IBM describes this process as filling a golf hole from a flying aircraft! One of the major stumbling blocks is the huge amount of heat generated when transistors are so closely packed. Witness the huge heat sink and fan combination that dissipate heat emitted by processors. Further, at atomic scales, it becomes difficult

What's What?

Problem: Chip manufacturers are finding it difficult to extend Moore's Law. Power dissipation, current leakage and manufacturing costs are preventing them from moving forward.

Solution: Developments such as high-k/metal, self-assembly, carbon nanotubes, strained silicon and new lithography techniques promise to keep the torch lighted.

Bottom line: Moore's Law will be valid at least for another 10 years. It may slow down after that.